

1 Path analysis

Path	B	SE	z	p	Std. β	95% CI	R ²
x1 $\rightarrow$ x2	0.30	0.06	5.23	<.001	.30	[.19, .40]	.09
x2 $\rightarrow$ x3	0.22	0.05	4.05	<.001	.23	[.12, .34]	.24
x1 $\rightarrow$ x3	0.36	0.05	7.29	<.001	.37	[.28, .47]	.24

Path	Estimate	SE	boot 95% CI	p
x1 $\rightarrow$ x2 $\rightarrow$ x3	0.07	0.02	[0.03, 0.11]	.001

x1

$\beta = .30$

Path analysis estimates a system of regressions among observed variables at once, letting one variable be both an outcome and a predictor (mediation). Structural paths: each B is the unstandardized effect, $\text{Std. } \beta$ the standardized effect, with z , p , and a 95% CI; R^2 is the variance explained in each endogenous variable. Indirect effects: the product of the paths through a mediator (e.g. $X \rightarrow M \rightarrow Y$), judged by its bootstrap 95% CI — an interval excluding 0 indicates mediation. When $df = 0$ the model is saturated (it reproduces the data exactly), so fit indices are not informative and are omitted; only with extra constraints ($df > 0$) do global fit indices apply, and even then read RMSEA cautiously at small df / small N . All thresholds are heuristics, not pass/fail gates.

A path model fit to the observed variables; the indirect effect of X on Y through M was significant, bootstrap 95% CI excluding 0.

Statistical basis: Path model estimation (lavaan::sem, observed variables only) (Rosseel, Y. (2012). "lavaan: An R Package for Structural Equation Modeling." *Journal of Statistical Software*, 48(2), 1-36.); RMSEA interpreted cautiously at small df / small N for over-identified models (Kenny, D. A., Kaniskan, B., McCoach, D. B. (2015). "The performance of RMSEA in models with small degrees of freedom." *Sociological Methods & Research*, 44(3), 486-507.); Bootstrap percentile CIs for indirect (mediated) effects (MacKinnon, D. P., Lockwood, C. M., Williams, J. (2004). "Confidence limits for the indirect effect: Distribution of the product and resampling methods." *Multivariate Behavioral Research*, 39(1), 99-128.).